

Estrogen levels in nipple aspirate fluid and serum during a randomized soy trial

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Background Based on the hypothesized protective effect, we examined the effect of soy foods on estrogens in nipple aspirate fluid (NAF) and serum, possible indicators of breast cancer risk. **Methods** In a cross-over design, we randomized 96 women who produced ≥ 10 μ L NAF to a high- or low-soy diet for 6-months. During the high-soy diet, participants consumed 2 soy servings of soy milk, tofu, or soy nuts (approximately 50 mg of isoflavones/day); during the low-soy diet, they maintained their usual diet. Six NAF samples were obtained using a FirstCyte© Aspirator. Estradiol (E2) and estrone sulfate (E1S) were assessed in NAF and estrone (E1) in serum only using highly sensitive radioimmunoassays. Mixed-effects regression models accounting for repeated measures and left-censoring limits were applied. **Results** Mean E2 and E1S were lower during the high-soy than the low-soy diet (113 vs. 313 pg/mL and 46 vs. 68 ng/mL, respectively) without reaching significance ($p=0.07$); the interaction between group and diet and was not significant. There was no effect of the soy treatment on serum E2 ($p=0.76$), E1 ($p=0.86$), or E1S ($p=0.56$). Within individuals, NAF and serum levels of E2 ($r_s=0.37$; $p<0.001$) but not E1S ($r_s=0.004$; $p=0.97$) were correlated. E2 and E1S in NAF and serum were strongly associated ($r_s=0.78$ and $r_s=0.48$; $p<0.001$). **Conclusions** Soy foods in amounts consumed by Asians did not significantly modify estrogen levels in NAF and serum. **Impact** The trend towards lower estrogens in NAF during the high-soy diet counters concerns about adverse effects of soy foods on breast cancer risk.